

A/B Test Analysis: Ad Campaign Effectiveness

Personal Data Analytics Project | Public marketing A/B dataset (588,101 users) | Tools: Python (pandas, statsmodels, matplotlib), Google Colab | Role: Data Analyst (self-directed)

1. Project Overview

A company showed a real advertisement to one group of users (**ad**) and a neutral public service announcement to a control group (**PSA**). This project tests whether the ad increased the share of users who bought the product, measures how large the effect is, and looks for the conditions under which the ad works best.

2. Business Questions

- Did users who saw the ad convert at a higher rate than the PSA control?
- How large is the lift, and is it more than random chance?
- Does the effect depend on how many ads a user saw?
- Does timing (day of week and hour) matter?

3. Data and Checks

Public dataset with one row per user: test group, whether the user converted, total ads seen, the day with the most ad exposure and the hour with the most ad exposure.

Check	Result
Users	588,101 (ad 564,577 = 96%; PSA 23,524 = 4%)
Missing values	None in any column
Duplicate users	None
Group balance	Very unequal; the PSA control is small, so its estimates are less precise

4. Method

- **Metric:** conversion rate = users who bought / users in the group.
- **H0:** the ad and PSA groups have the same conversion rate. **H1:** they differ. Significance level 0.05.
- **Test:** two-proportion z-test, plus a 95% confidence interval for the difference in rates.
- **Follow-up analysis:** conversion by number of ads seen (five bands), by day and by hour.

5. Main Result

Group	Users	Converted	Conversion rate
Ad	564,577	14,423	2.55%
PSA (control)	23,524	420	1.79%

The ad group converted **0.77 percentage points** higher, a relative lift of about **43%**. The z-test gives **$z = 7.37$** , **$p = 1.7 \times 10^{-13}$** , and the 95% confidence interval for the gain is **0.59 to 0.94 percentage points**. The interval excludes zero, so the difference is very unlikely to be chance. Across the 564,577 users in the ad group, this gain is roughly 4,300 extra purchases, assuming the ad caused the difference.

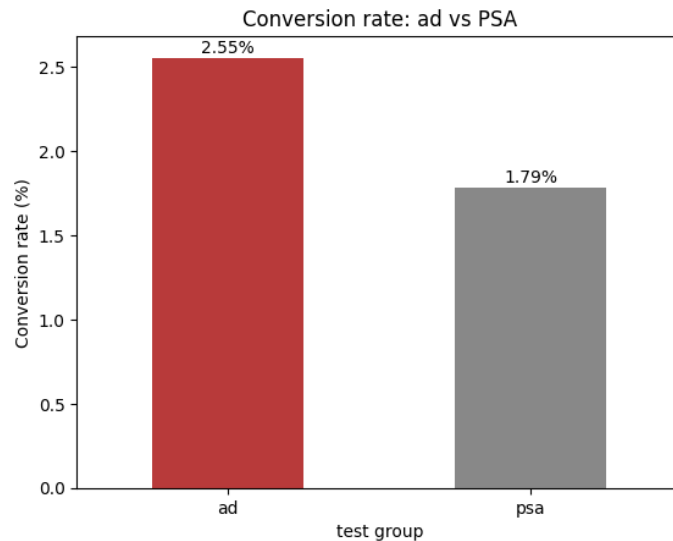


Fig. 1: Conversion rate, ad vs PSA.

6. Does the effect depend on how many ads were seen?

Ads seen	Ad rate	PSA rate	Gap (points)	Users (ad / PSA)
1-10	0.33%	0.40%	-0.07	249,499 / 11,276
11-50	1.89%	1.58%	0.31	248,875 / 9,385
51-100	11.63%	5.77%	5.86	44,149 / 1,853
101-500	17.13%	11.91%	5.22	21,488 / 991
500+	17.49%	10.53%	6.96	566 / 19

The overall lift comes almost entirely from users who saw many ads. At 1 to 10 ads the ad shows no benefit over the PSA (the difference is within chance), at 11 to 50 ads the gain is small, and from 51 ads upward the ad group converts far higher. The 500+ band has only 19 PSA users, so its gap should not be relied on.

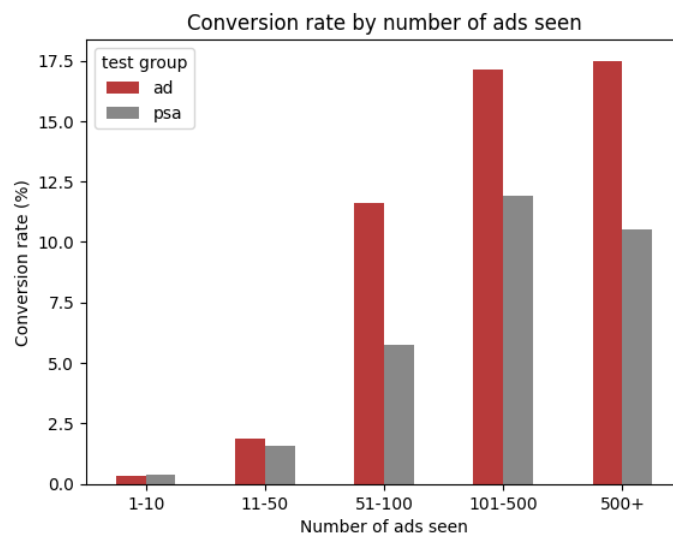


Fig. 2: Conversion rate by number of ads seen.

7. Timing: day and hour

The ad group is ahead of the PSA group on every day of the week. Conversion in the ad group is highest when most ads were seen on Monday (3.32%) and Tuesday (3.04%), and the gap is smallest on Thursday (0.14 points). By hour, conversion rises through the day and peaks in the mid-afternoon (about 3.1% around 16:00), and the ad line sits above the PSA line at almost every hour. The PSA line is erratic overnight, with 0% at several early hours, because those hours have very few PSA users.

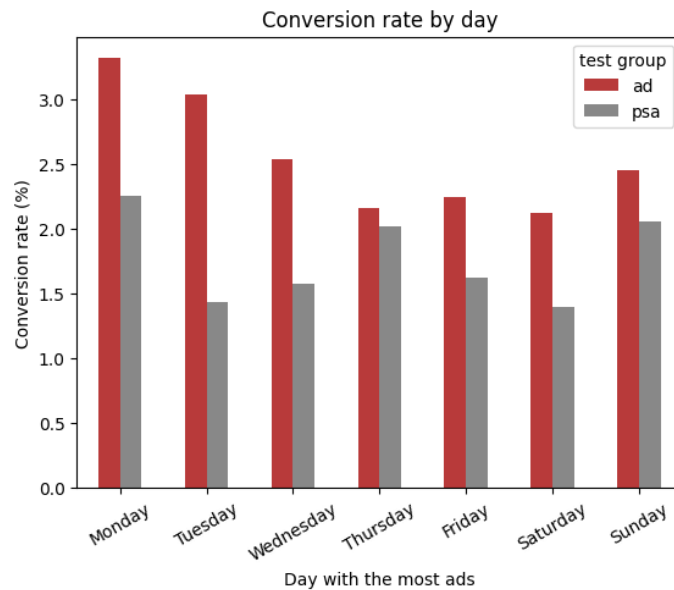


Fig. 3: Conversion rate by day with the most ads.

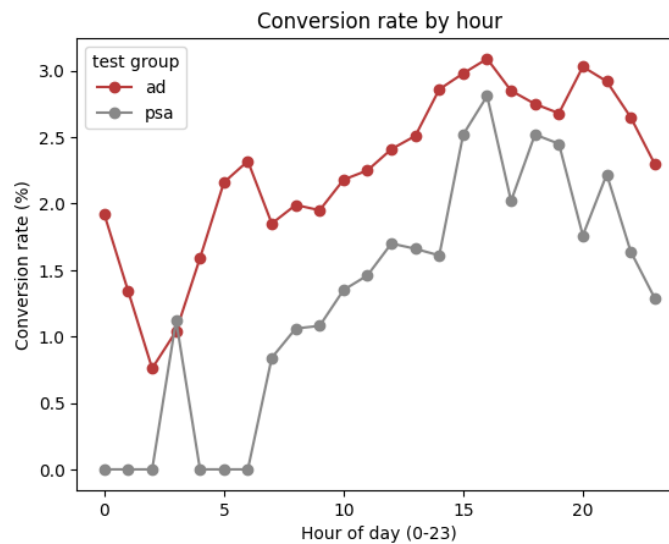


Fig. 4: Conversion rate by hour. Overnight PSA values rest on very small samples.

8. Key Insights

- The ad works overall: 2.55% vs 1.79% conversion (+43% relative), statistically significant ($p < 0.001$), with a gain of 0.59 to 0.94 points.
- The effect is concentrated among heavily exposed users. With 10 or fewer ads there is no lift; with 51 or more the gap is more than 5 points.

- The ad outperforms the PSA on every day of the week, most clearly early in the week.
- Conversion is highest in the afternoon and evening for both groups.
- Conversion also rises steeply with ad count in the PSA group, so heavy exposure may partly reflect users who were already more engaged.

9. Recommendations

- Keep running the ad; the evidence suggests it increases conversion.
- Test ad frequency directly (for example, a randomized frequency cap), because the gain appears only at higher exposure and this data cannot show whether more ads cause more purchases.
- Weight delivery toward afternoon and evening hours, and review the weaker Thursday results.
- Before scaling spend, calculate cost per extra purchase, which this dataset cannot show because it has no cost or revenue data.

10. Limitations

- The groups are very unequal (96% ad, 4% PSA), so estimates for the control group, and especially for small segments, are noisy.
- It is a public dataset and the assignment process cannot be verified. The test is only valid if users were randomly assigned.
- Total ads seen is measured after exposure and differs between users, so the exposure breakdown describes patterns and does not prove that more ads cause more conversions.
- The day and hour columns record when a user saw the most ads, not when they bought.
- Only conversion was tested, with no revenue, cost or long-term value, and many segment comparisons raise the chance of a false pattern.

11. Skills Applied

Hypothesis testing, two-proportion z-test, confidence intervals, A/B test design, segmentation analysis, Python (pandas, statsmodels, matplotlib), data validation, communicating results and limits.

12. Conclusion

The ad increased conversion from 1.79% to 2.55%, a result that is statistically significant and practically large. The deeper analysis adds the more useful business point: the benefit is concentrated among users who see the ad many times, so the next step is a controlled test of ad frequency and a calculation of cost per extra purchase.